Akilan Jayakumar Deepa

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Executive Summary

AI Master's student at Brunel University London with a strong background in computer science and applied AI. Skilled in Python, OpenCV, and LLMs. Experienced in developing intelligent, user-focused systems in chatbot, media processing, and civic automation domains. Experienced in writing clean, maintainable, production-oriented code through real-world projects. Actively seeking opportunities to bring scalable, production-ready AI solutions to ambitious teams.

Skills

Languages: C/C++, Python, JavaScript, SQL
AI/ML: TensorFlow, Scikit-learn, PyTorch, OpenCV, MoviePy, LM Studio, Prompt Engineering.
Web & Full-Stack: ReactJS, Node.js, HTML5, CSS3, Webflow.
Databases: MySQL, Firebase.
Data Visualisation: Power BI, Tableau.
Tools & Platforms: GitHub, VS Code, Figma, Microsoft Office Suite.
Concepts: Object-Oriented Programming (OOP), RESTful APIs, Agile Development.

Project Works

AI-Powered Customizable Chatbot

- Led Development of an offline chatbot using a pre-trained LLM integrated with LM Studio.
- Added a GPU utilization control feature, enabling users to select GPU load for faster response generation. Enabled multi-domain conversations and chat log downloads for reusability.
- Delivered scalable AI assistance offline with optimized GPU performance and user control.
  Stack: Python, LM Studio

Media Processing & Real-Time Editing System

- Developed a media editor supporting live image adjustments and video operations (merge, trim, resolution).
- Led the project including architecture, logic, and UI implementation. Delivered optimized real-time performance using Python libraries. Streamlined media editing workflows, reducing manual editing time.
  Stack: Python, OpenCV, MoviePy.

Public Issue Reporting System (In Development)

- Designed a full-stack web app for citizens to report local issues with geo-tagging. Developed modules for complaint submission, officer assignment, and live tracking.
- Planning AI integration for auto-categorization, urgency detection, and heatmap analytics. Simplified public reporting and improved case tracking efficiency.
  Stack: React, Node.js, MySQL/Firebase

Education

Brunel University London, United Kingdom Sep 2025 - Present
MSc Artificial Intelligence
Relevant Coursework: Advanced Artificial Intelligence, Machine Learning Applications, Deep Learning and Neural Networks, Modern Data, Quantitative Data Analysis (QDA).

SRM Institute of Science and Technology, Chennai, India Aug 2021 - May 2025
B.Tech Computer Science and Engineering (Specialisation in AI & ML) ***CGPA: 8.80 / 10***
Relevant Coursework: Object Oriented Programming, Data Structures and Algorithms, AI & ML, Database Management Systems (DBMS), Web Application Development, Data Analytics and Visualisation, Human-Computer Interaction (HCI).

Awards & Certificates

- Completed *Generative AI with Large Language Models* course by DeepLearning.AI on Coursera, gaining hands-on experience in LLMs, prompt engineering, and applied machine learning using PyTorch. (2024)
- Completed *Foundations of User Experience (UX) Design* by Google on Coursera, focusing on user-centric design, accessibility, and design sprint methodology. (2024)